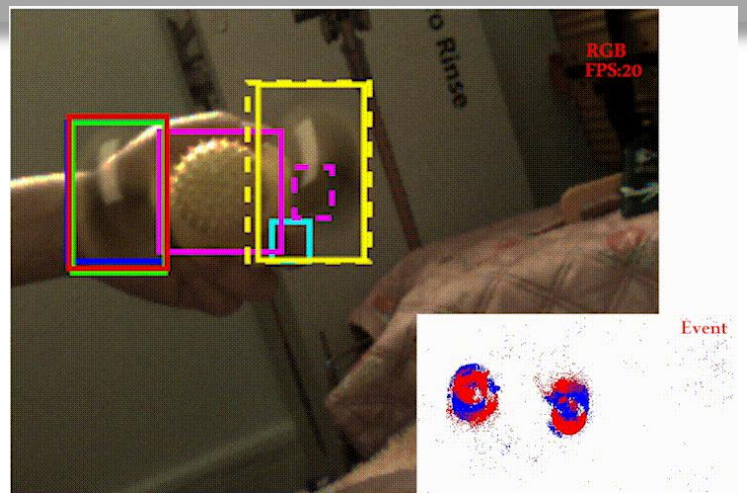
CS5187: Vision & Image

香港城市大學（東莞）
City University of Hong Kong
(Dongguan)

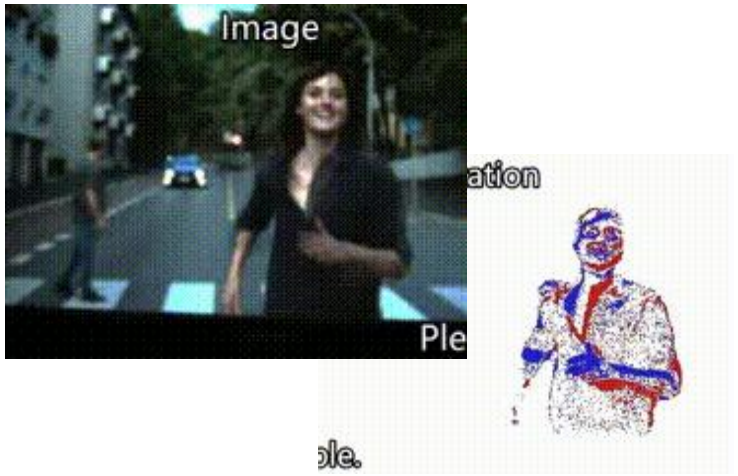
Instructor: Zhu Zhiyu











香港城市大學（東莞）
City University of Hong Kong
(Dongguan)

Instructor

- Zhu Zhiyu (zhiyu.zhu@cityu.edu.hk)
- Research interests:
  - Computer vision and computational imaging
  - Machine learning (Generative models)

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City University of Hong Kong
(Dongguan)

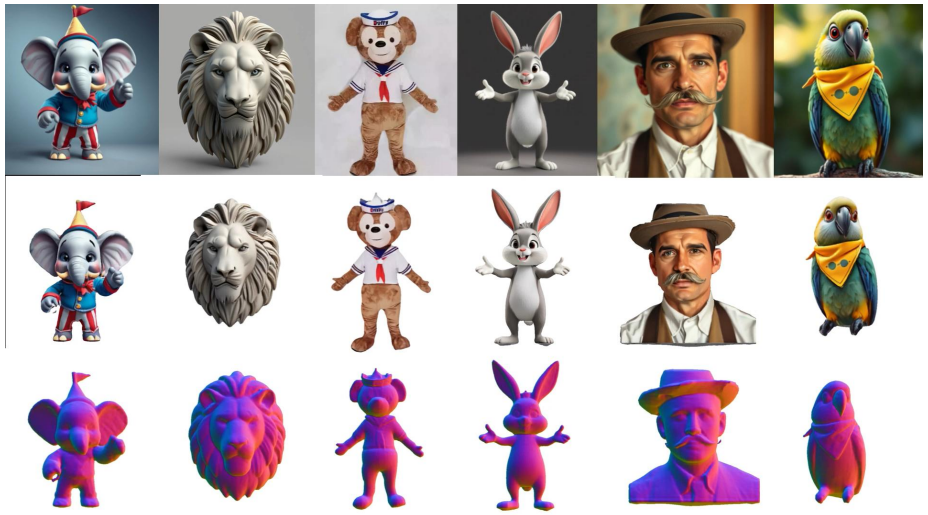
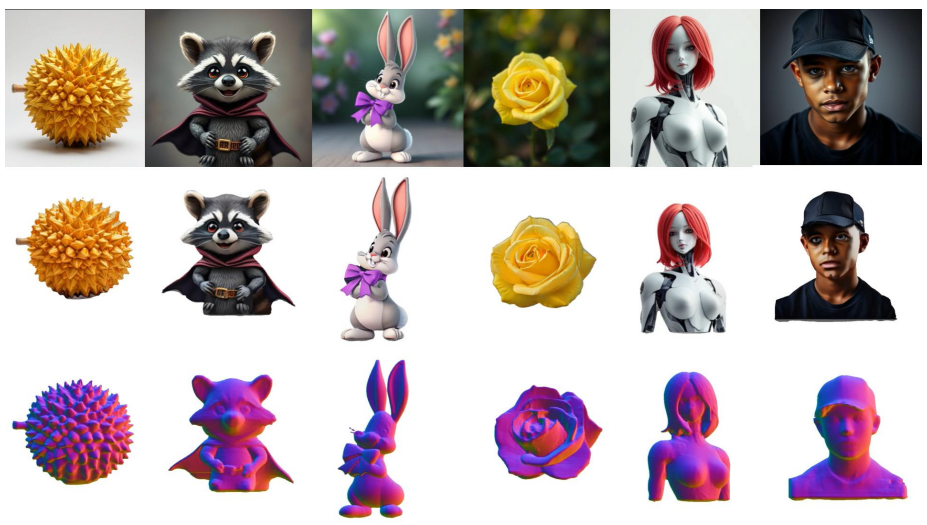
My research work

- Novel view Synthesis from single static images



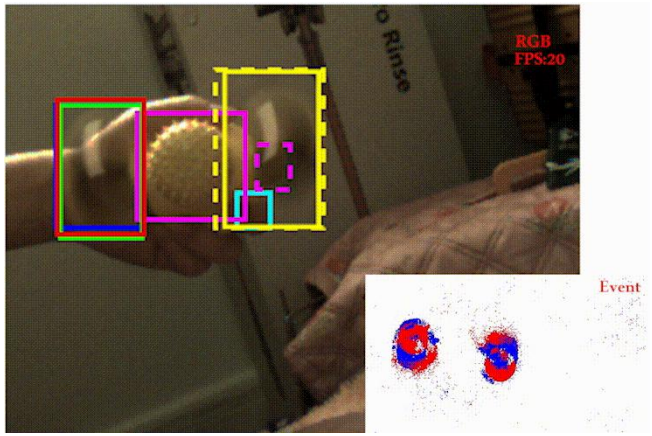


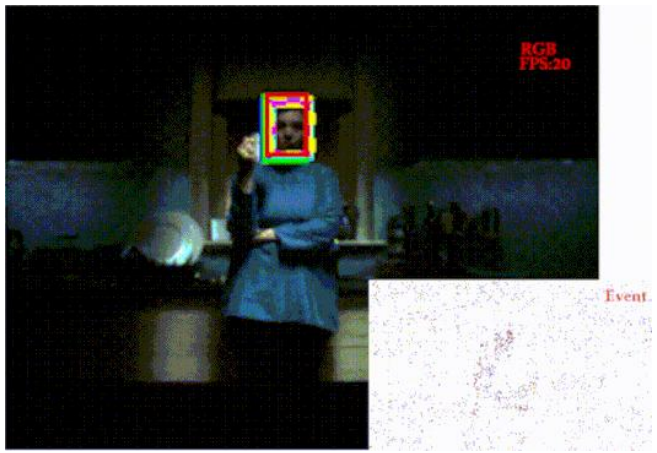
- 3D content generation

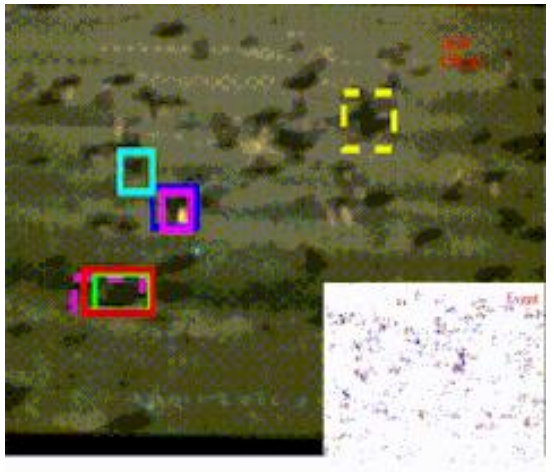



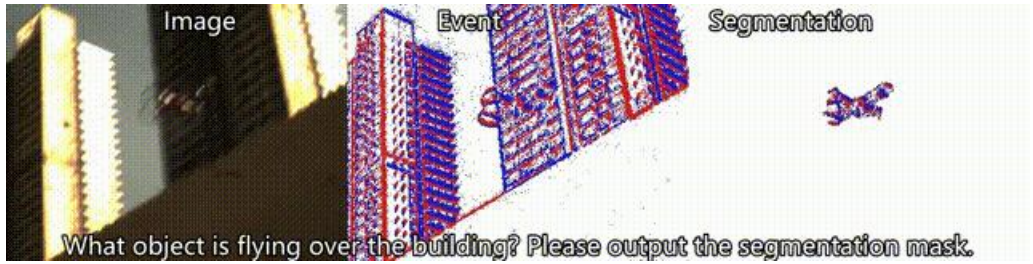
My research work

- Object tracking & detection





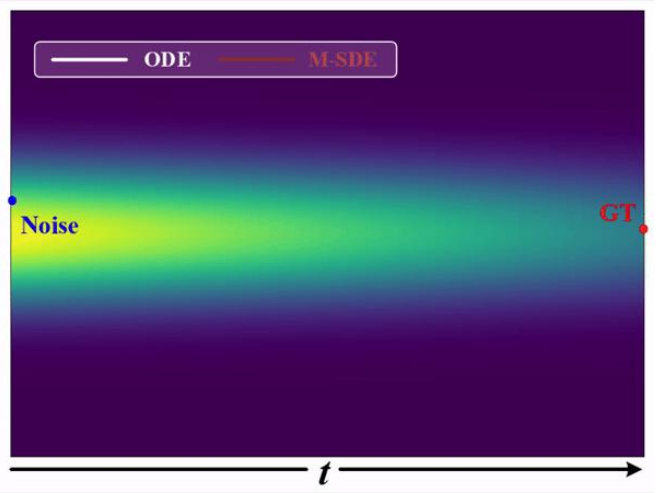


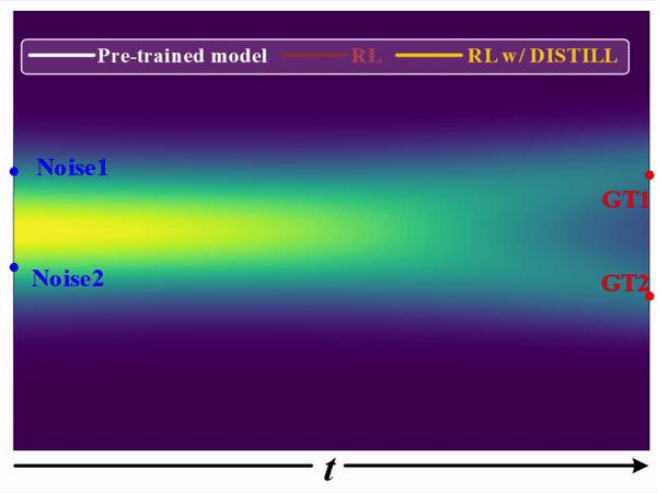


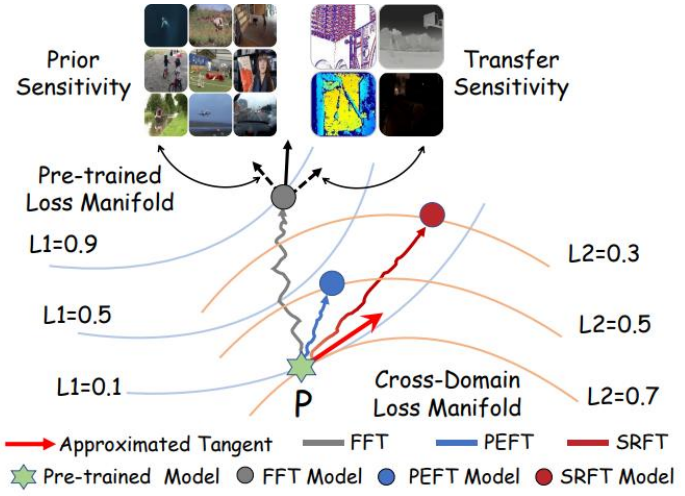


My research work

- Generative models & network learning theory







said

Today

1. What is computer vision?
2. Why study computer vision?
3. Course overview
4. Images & image filtering [time permitting]

Every image tells a story



- Goal of computer vision: perceive the "story" behind the picture
- Compute properties of the world
  - 3D shape
  - Names of people or objects
  - What happened?

The goal of computer vision



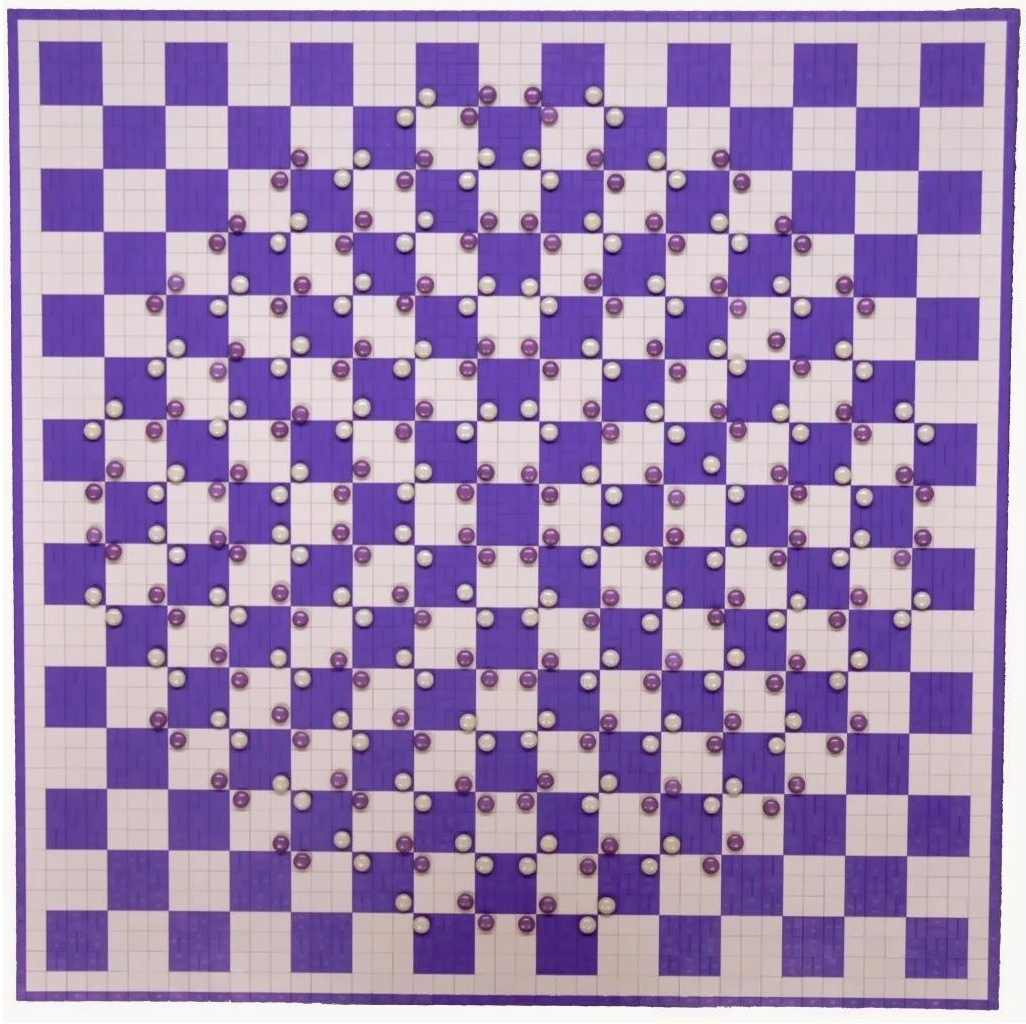
0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

Can computers match human perception?



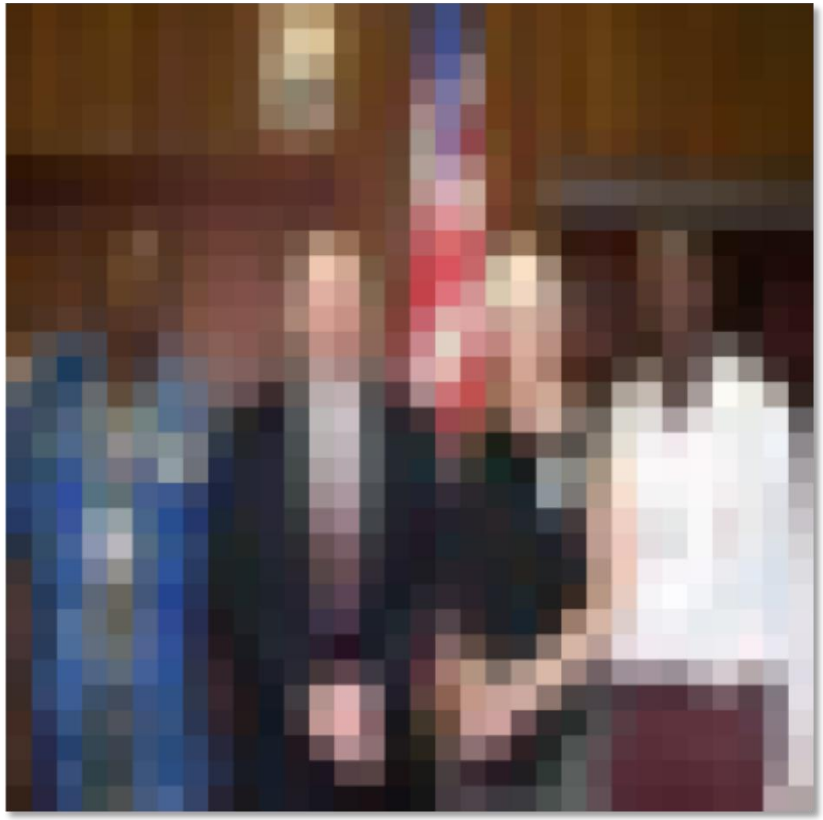
- Yes and no
  - humans are better at “hard” things, are more robust, and make inferences quickly and cheaply
- But huge progress
  - Accelerating in the last 10 years due to deep learning, large vision-language models
  - What is considered “hard” keeps changing

Human perception has its shortcomings



https://twitter.com/pickover/status/1460275132958662657/

But humans can tell a lot about a scene from a little information…



Source: “80 million tiny images” by Torralba, et al.

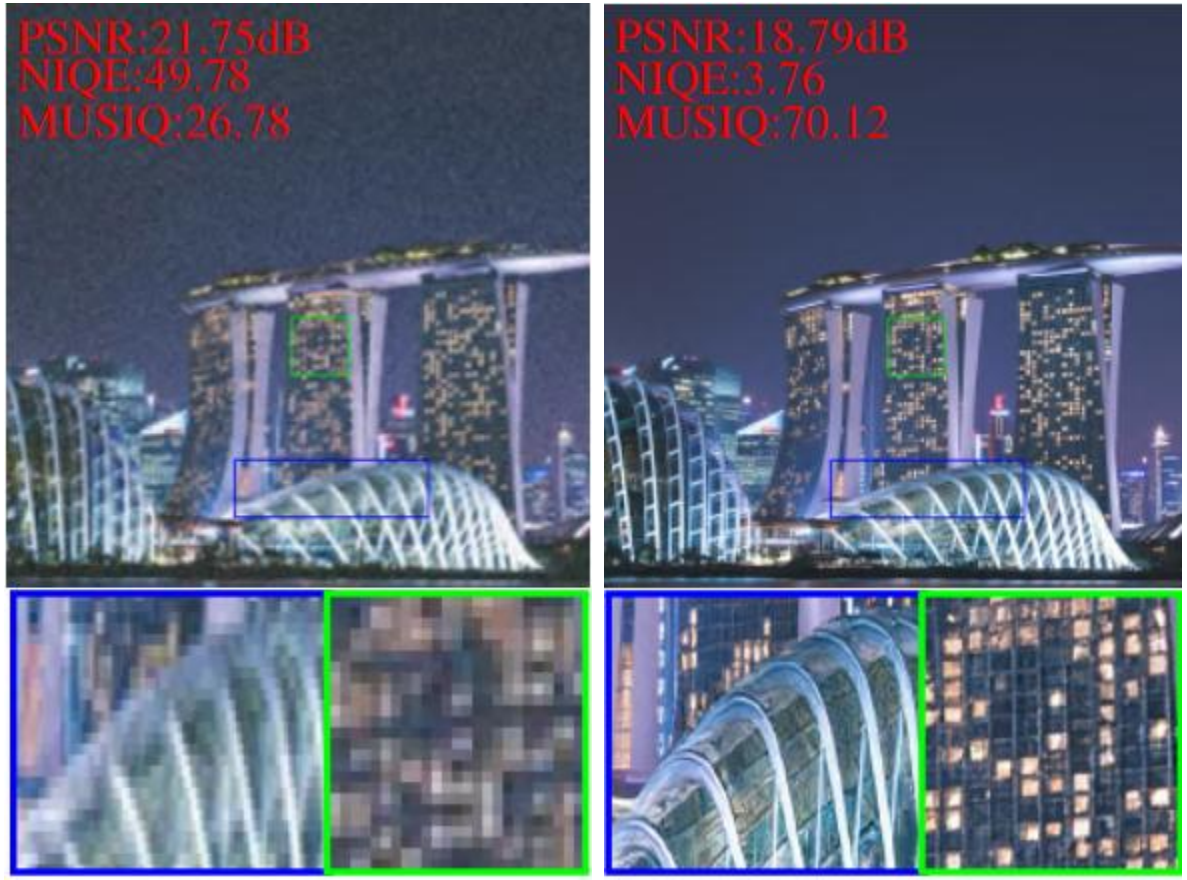


The goal of computer vision



The goal of computer vision

- “Enhance” images



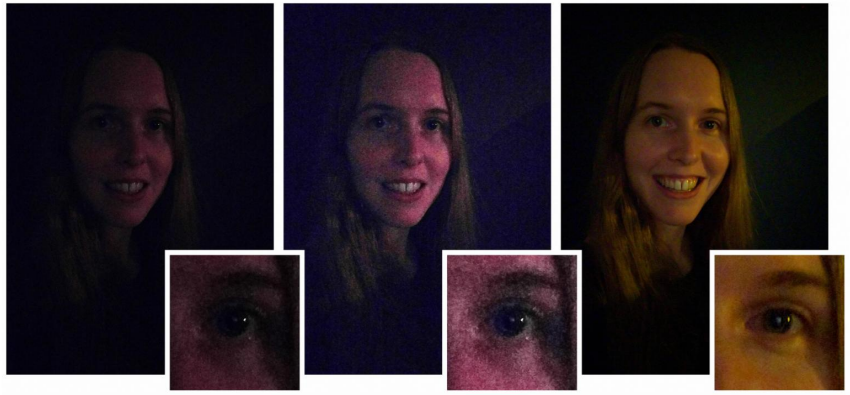


The goal of computer vision

- "Enhance" images



Super-resolution (source: 2d3)



Low-light photography
(credit: Hasinoff et al., SIGGRAPH ASIA 2016)



Depth of field on cell phone camera
(source: Google Research Blog)



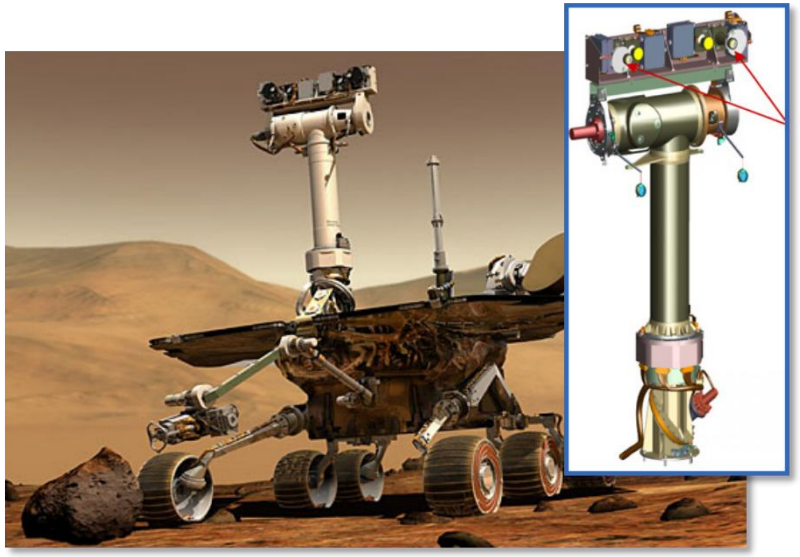
Removing objects
(Google Magic Eraser)

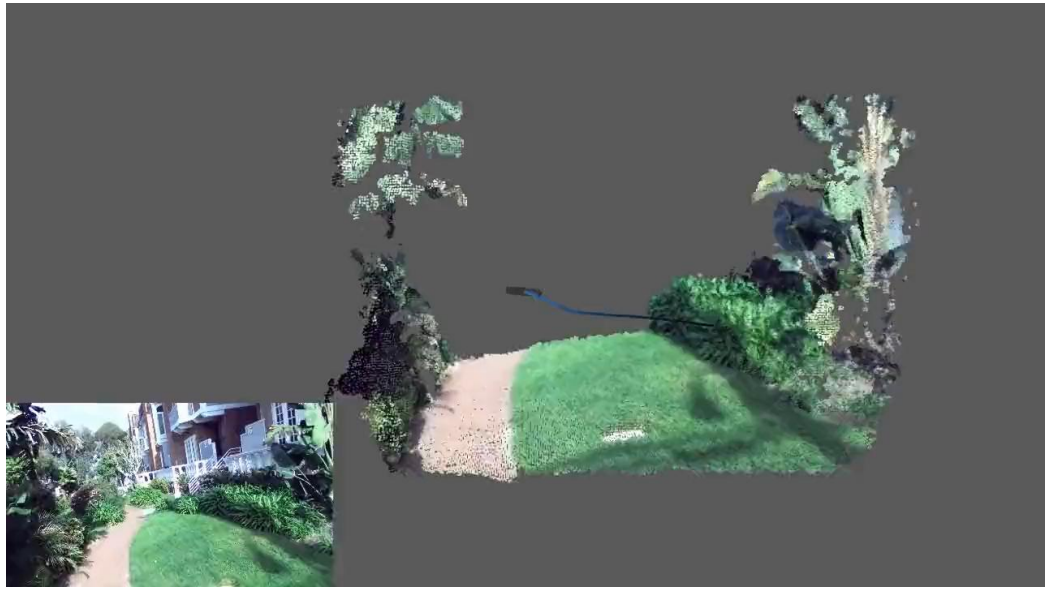
The goal of computer vision

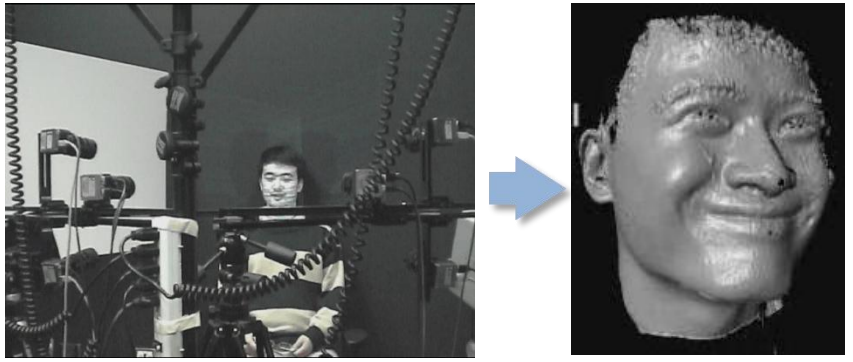
- Compute the 3D shape of the world



ZED 2i Camera

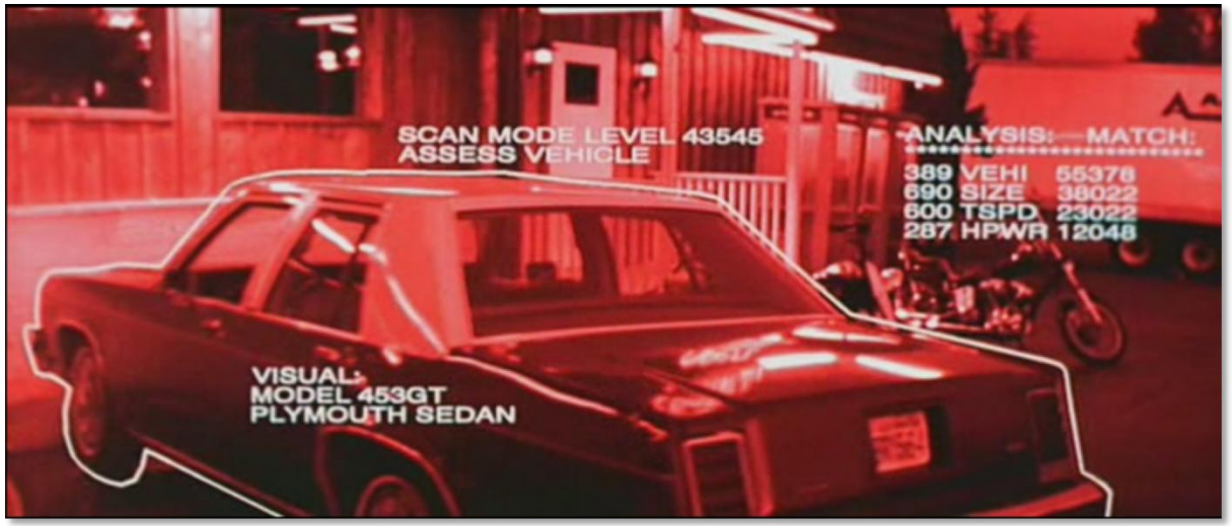






The goal of computer vision

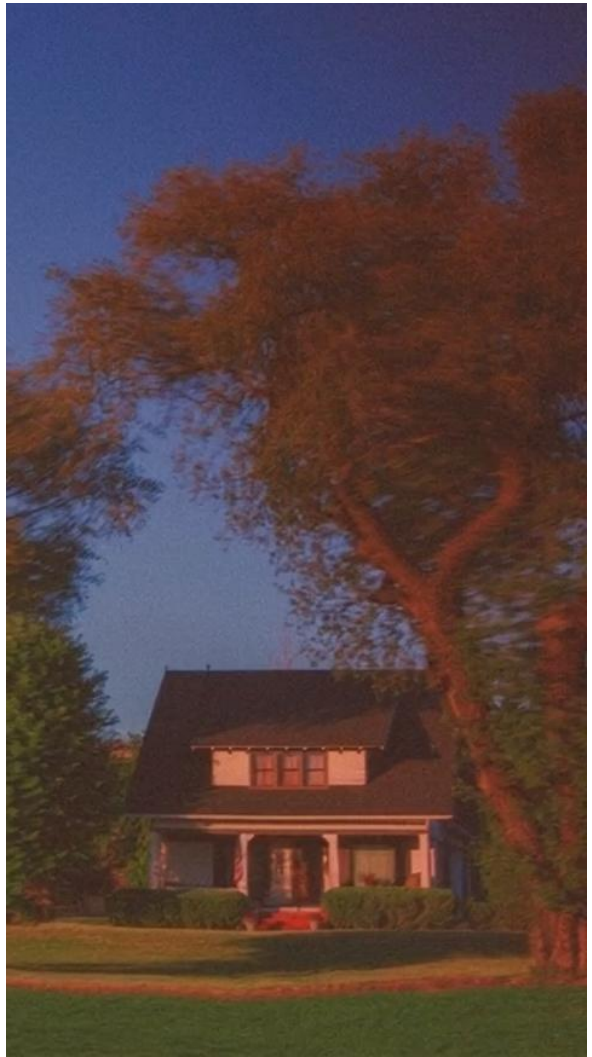
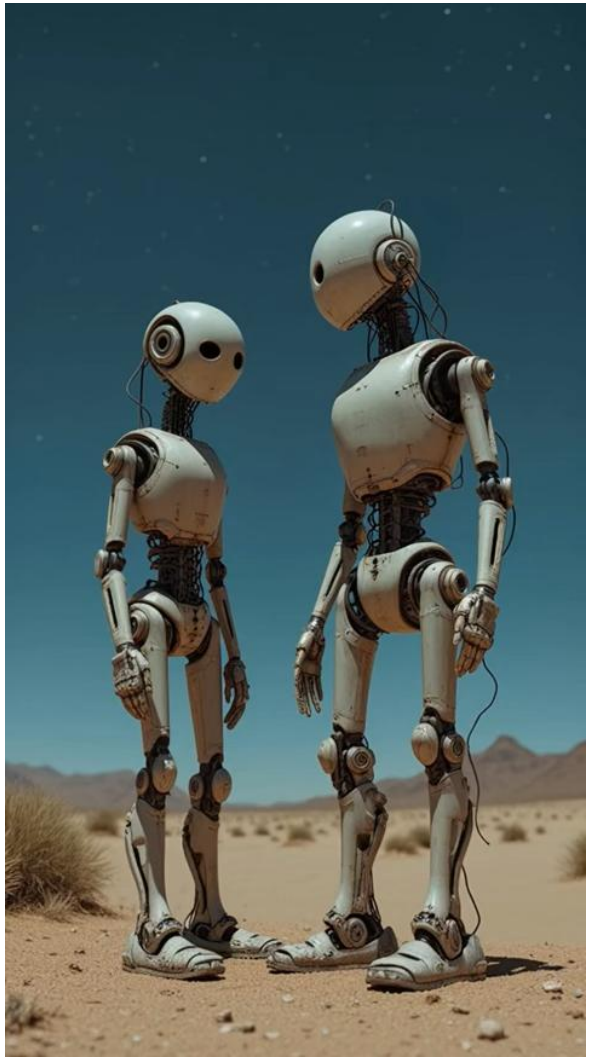
- Recognize objects and people



Terminator 2, 1991

The goal of computer vision

- Visual content generation.

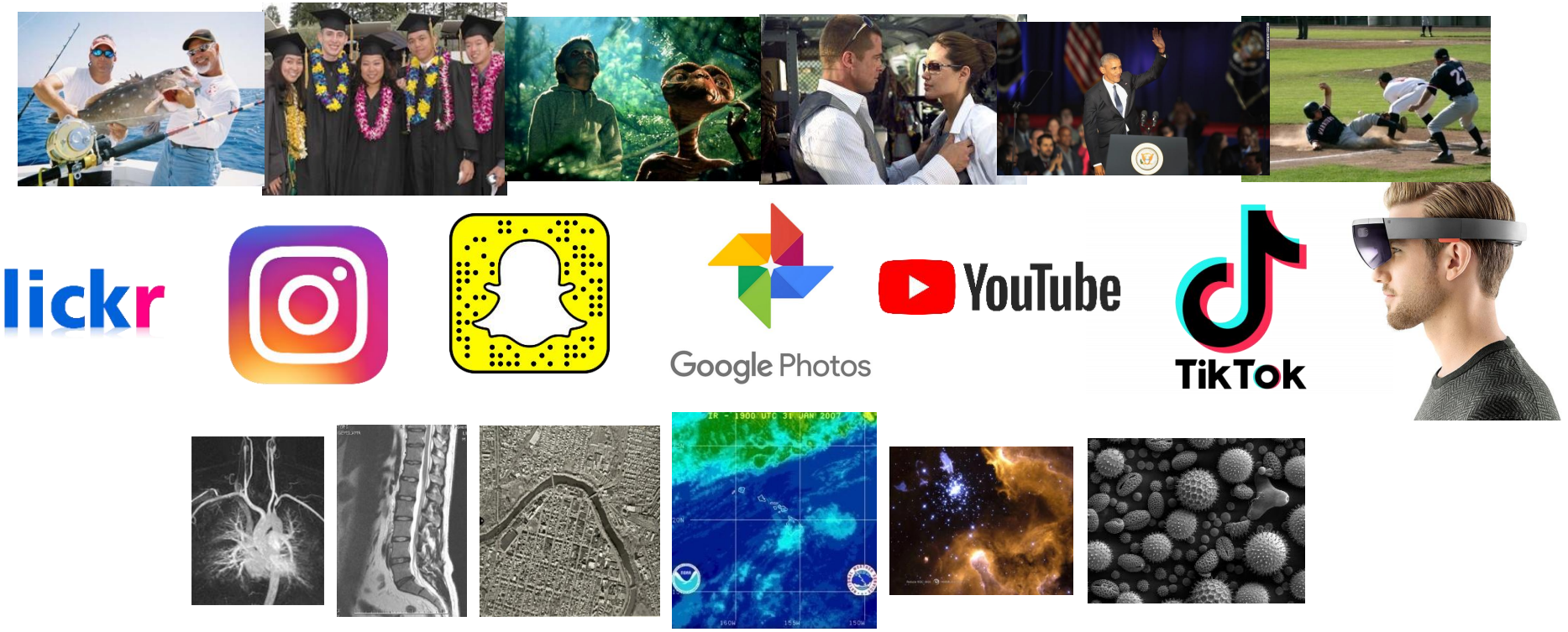





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Why study computer vision?

- Billions of images/videos captured per day



- Huge number of potential applications
- The next slides show the current state of the art

Optical character recognition (OCR)

- If you have a scanner, it probably came with OCR software



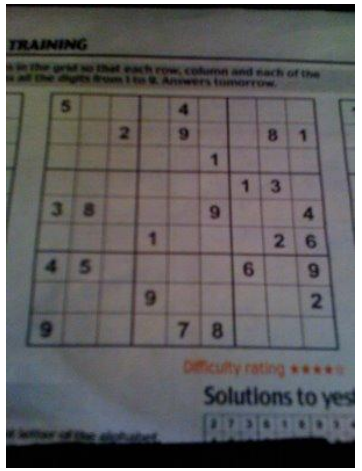
Digit recognition, AT&T labs (1990's)
http://yann.lecun.com/exdb/lenet/



License plate readers
http://en.wikipedia.org/wiki/Automatic_number_plate_recognition



Automatic check processing



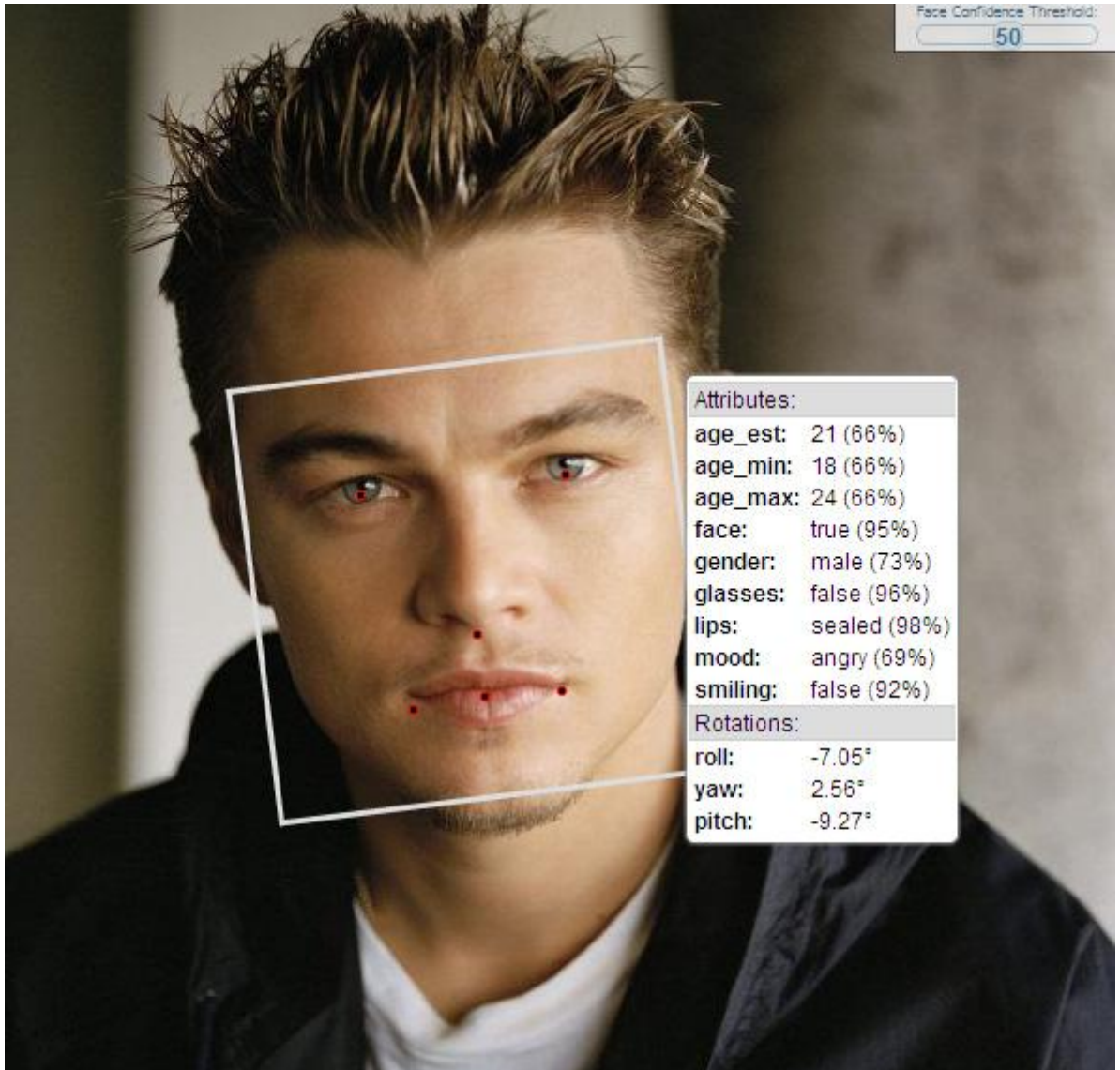
Sudoku grabber
http://sudokugrab.blogspot.com/

Face detection

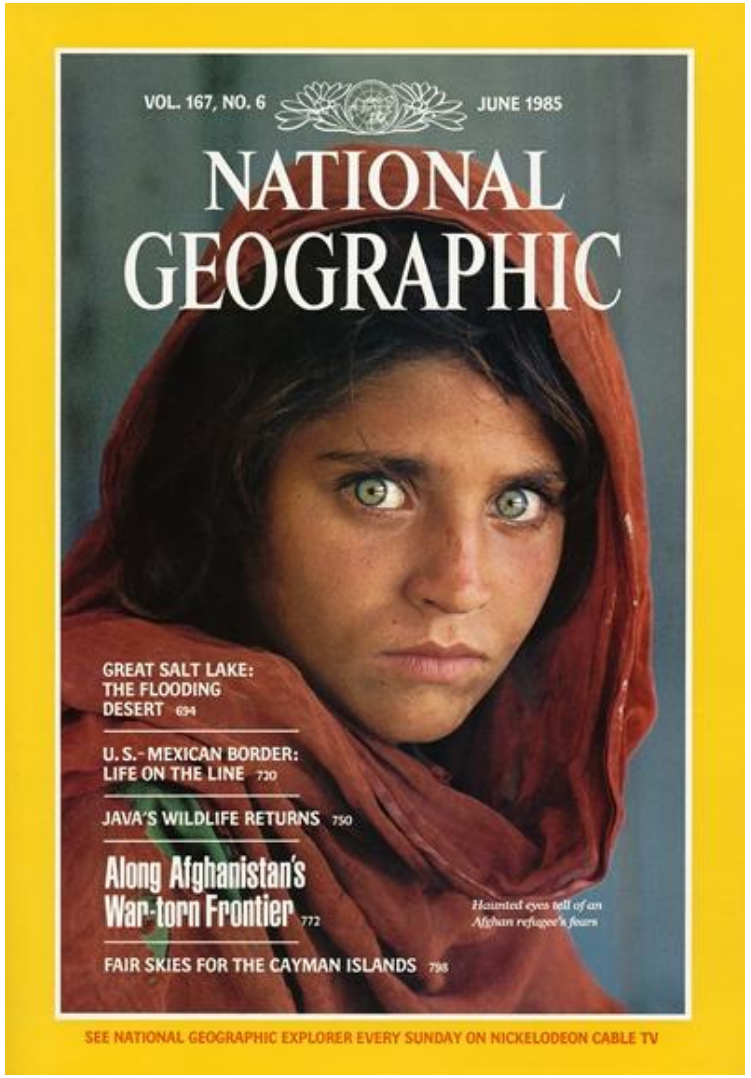


- Nearly all cameras detect faces in real time
  - (Why?)

Face analysis and recognition



Vision-based biometrics



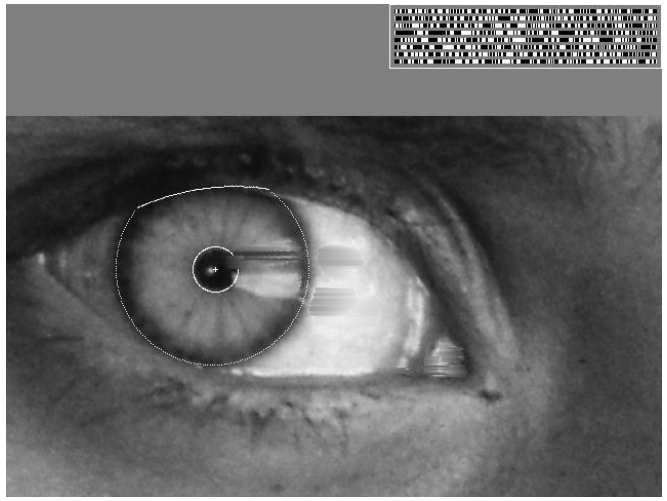
Who is she?

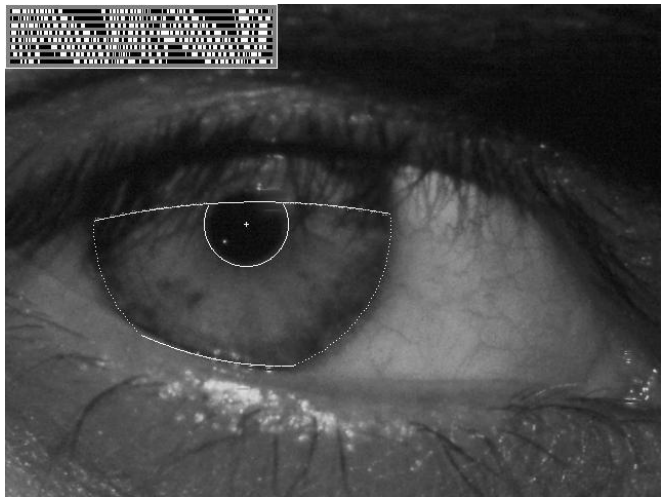
Source: S. Seitz

Vision-based biometrics



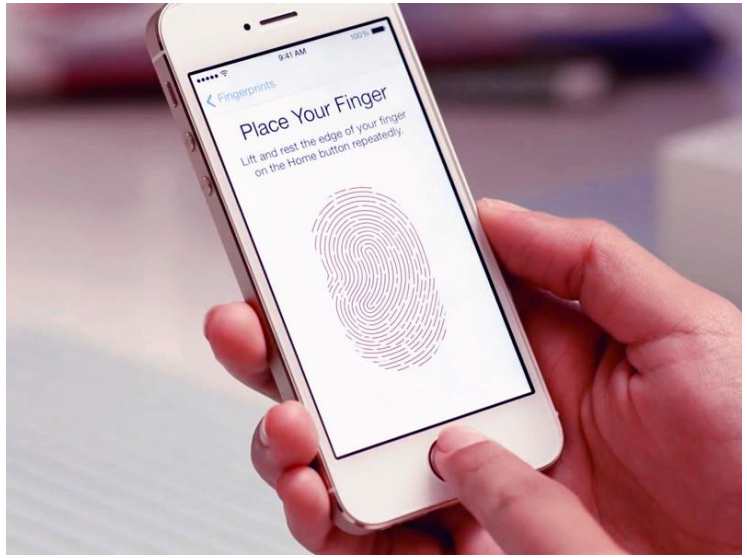
"How the Afghan Girl was Identified by Her Iris Patterns" Read the story





Source: S. Seitz

Login without a password



Fingerprint scanners on many new smartphones and other devices



Face unlock on Apple iPhone X
See also http://www.sensiblevision.com/

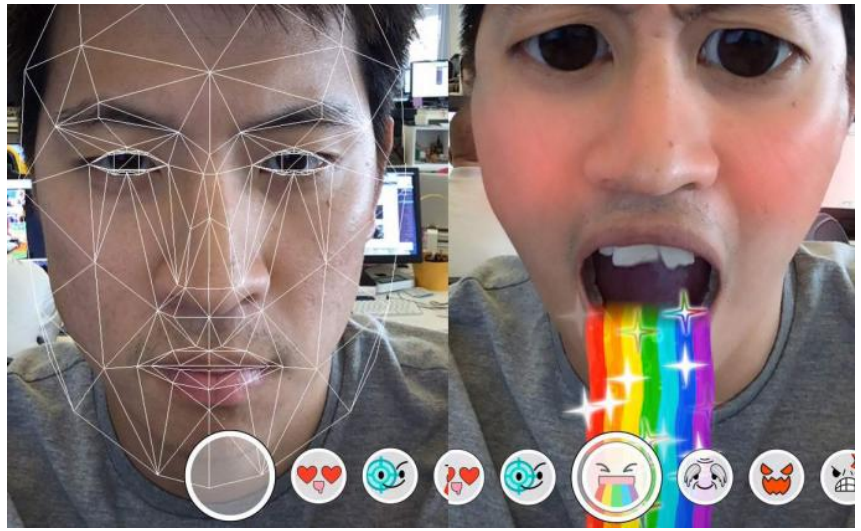
Special effects: motion capture



Pirates of the Carribean, Industrial Light and Magic

Source: S. Seitz

3D face tracking w/ consumer cameras



Snapchat Lenses

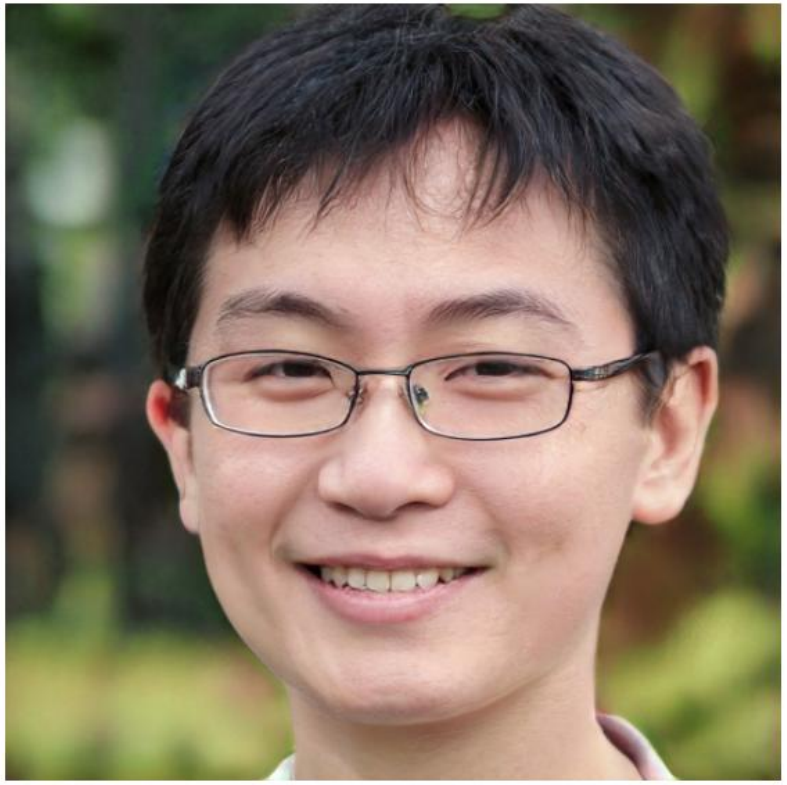


Face2Face system (Thies et al.)

Which face is real?

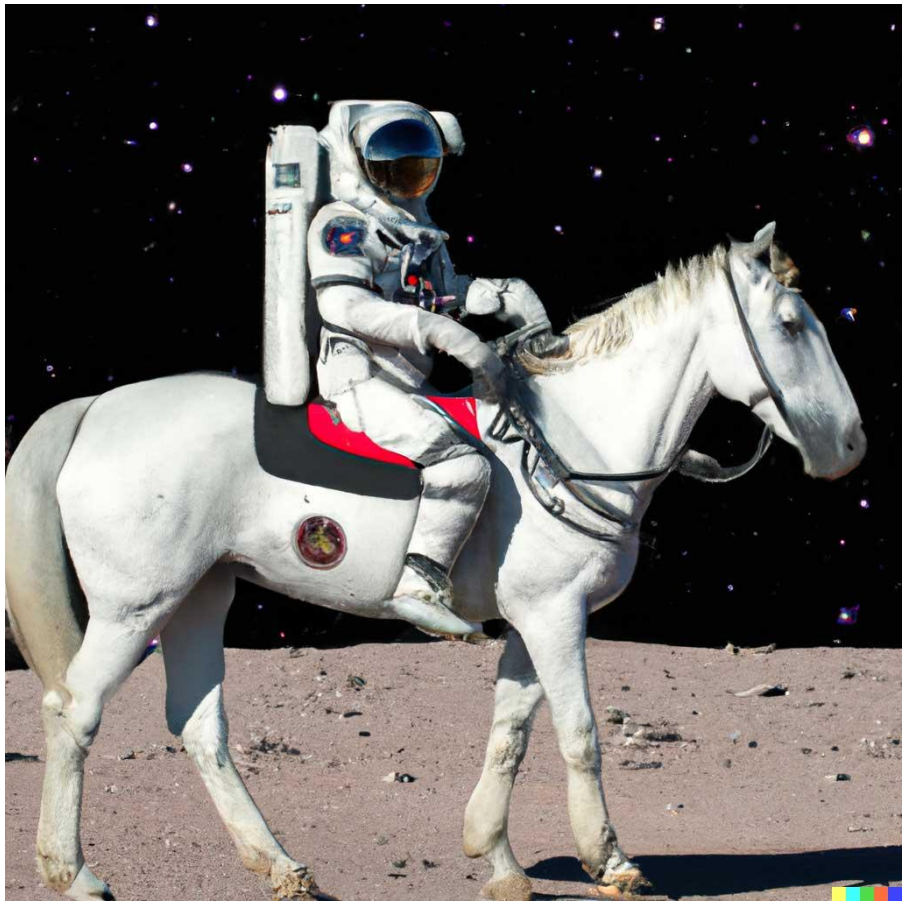
Click on the person who is real.





https://www.whichfaceisreal.com/

Image synthesis



"An astronaut riding a horse in a photorealistic style" – DALL-E 2



"A photo of a Corgi dog riding a bike in Times Square. It is wearing sunglasses and a beach hat" – Imagen

Sports



Sportvision first down line
Explanation on www.howstuffworks.com



Source: S. Seitz

Smart cars



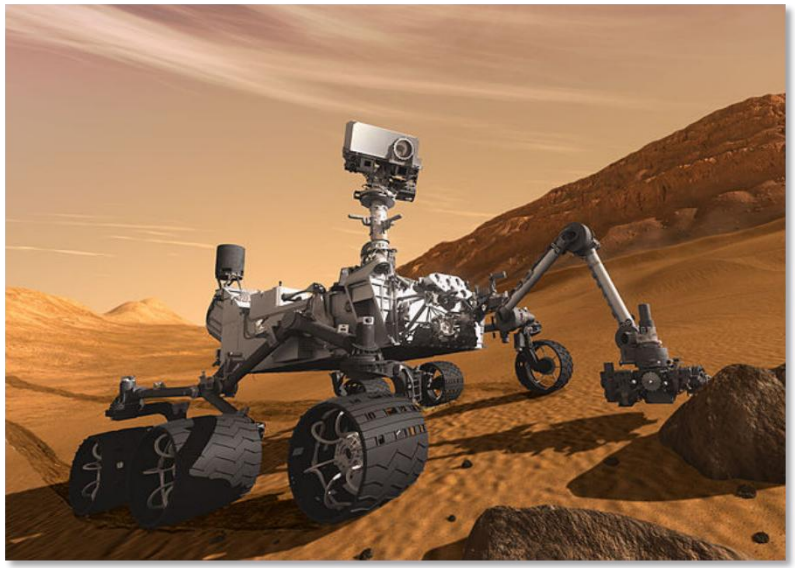
- Mobileye
- Tesla Autopilot
- Safety features in many cars

Self-driving cars



Waymo

Robotics



NASA's Mars Curiosity Rover
https://en.wikipedia.org/wiki/Curiosity_(rover)



Amazon Picking Challenge
http://www.robocup2016.org/en/events/amazon-picking-challenge/

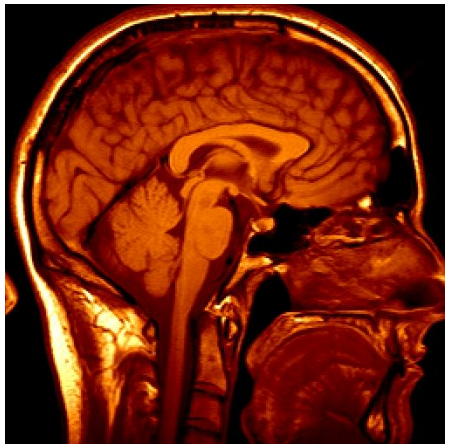


Amazon Prime Air

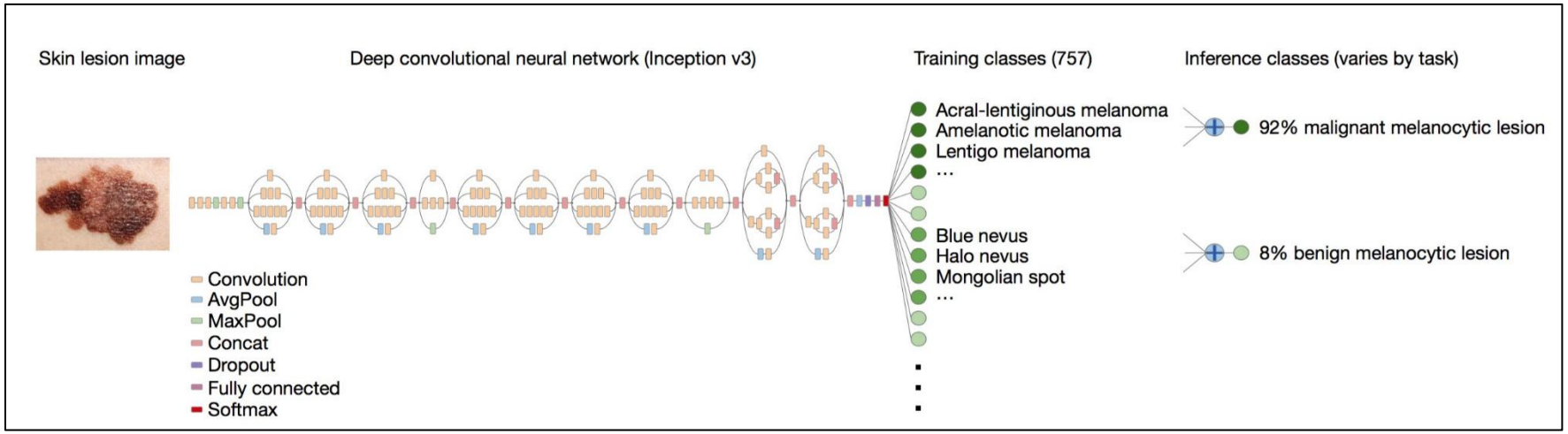


Amazon Scout

Medical imaging

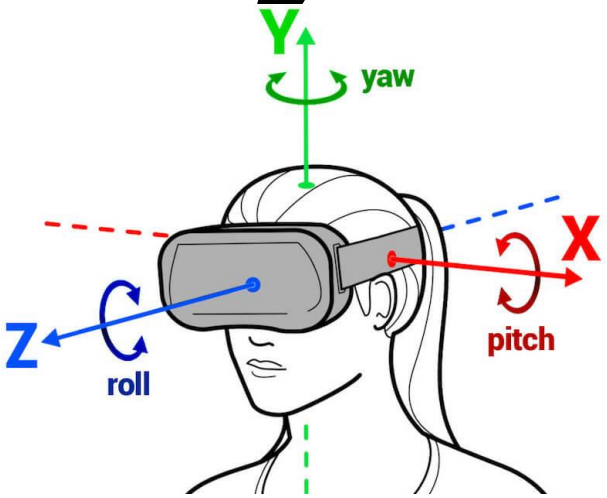


3D imaging
(MRI, CT)



Skin cancer classification with deep learning
https://cs.stanford.edu/people/esteva/nature/

Virtual & Augmented Reality



6DoF head tracking



Hand & body tracking



3D scene understanding



3D-360 video capture

Current state of the art

- You just saw many examples of current systems.
  - Many of these are less than 10 years old
- Computer vision is an active research area, and rapidly changing
  - Many new apps in the next 5 years
  - Deep learning and generative methods powering many modern applications
- Many startups across a dizzying array of areas
  - Generative AI, robotics, autonomous vehicles, medical imaging, construction, inspection, VR/AR, …

Why is computer vision difficult?



Viewpoint variation



Illumination



Scale

Why is computer vision difficult?



Intra-class variation



Motion (Source: S. Lazebnik)



Background clutter

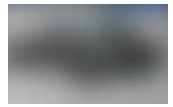


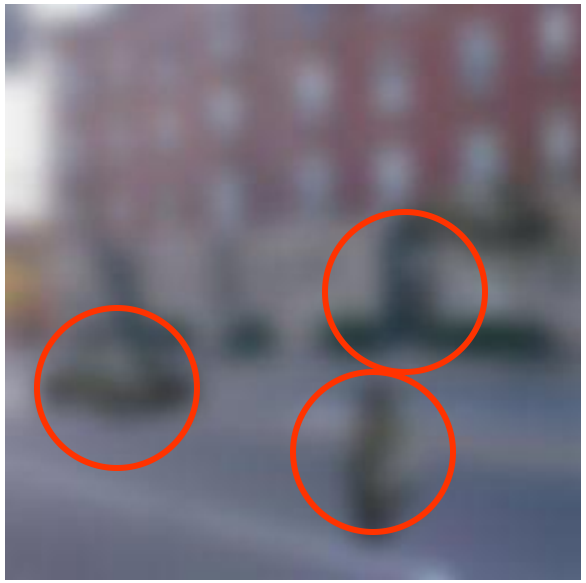
Occlusion

Challenges: local ambiguity











slide credit: Fei-Fei, Fergus & Torralba

But there are lots of visual cues we can use…

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Source: S. Lazebnik

CS5187: Vision and Image

- Project-based course whose goal is to teach you the fundamentals of computer vision – image processing, geometry, recognition – in a hands-on way

- This course covers **fundamentals**, **including mathematical fundamentals**. It is not a course specifically on deep learning or generative AI, though those topics will be covered.

Course requirements

- Prerequisites
  - Data structures
  - Good working knowledge of Python programming
  - Linear algebra
  - Vector calculus

- Course does ***not*** assume prior imaging experience
  - computer vision, image processing, graphics, etc.

Course overview (tentative)

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We generally have 13 classes.

3 Classes
20/01/2026 - 03/02/2026

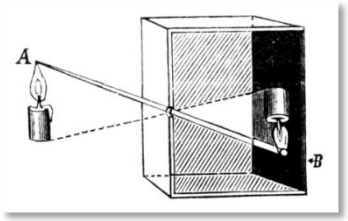
10 classes
03/03/2026 - 09/05/2026

	Empl ID	Name	Term	Class Nbr	Subject Area	Catalog Nbr	Description	Course Component	Week	Date	Weekday	Start Time
1	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	1	01/20/2026	Tues.	14:00
2	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	1	01/20/2026	Tues.	16:00
3	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	2	01/27/2026	Tues.	14:00
4	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	2	01/27/2026	Tues.	16:00
5	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	3	02/03/2026	Tues.	14:00
6	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	3	02/03/2026	Tues.	16:00
7	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	4	03/03/2026	Tues.	14:00
8	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	4	03/03/2026	Tues.	16:00
9	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	5	03/10/2026	Tues.	14:00
10	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	5	03/10/2026	Tues.	16:00
11	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	6	03/17/2026	Tues.	14:00
12	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	6	03/17/2026	Tues.	16:00
13	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	7	03/24/2026	Tues.	14:00
14	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	7	03/24/2026	Tues.	16:00
15	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	8	03/31/2026	Tues.	14:00
16	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	8	03/31/2026	Tues.	16:00
17	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	9	04/07/2026	Tues.	14:00
18	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	9	04/07/2026	Tues.	16:00
19	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	10	04/14/2026	Tues.	14:00
20	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	10	04/14/2026	Tues.	16:00
21	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	11	04/21/2026	Tues.	14:00
22	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	11	04/21/2026	Tues.	16:00
23	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	12	04/28/2026	Tues.	14:00
24	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	12	04/28/2026	Tues.	16:00
25	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	13	05/09/2026	Sat.	14:00
26	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	13	05/09/2026	Sat.	16:00

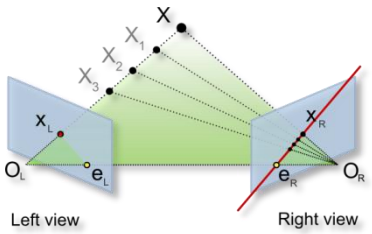
Course overview (tentative)

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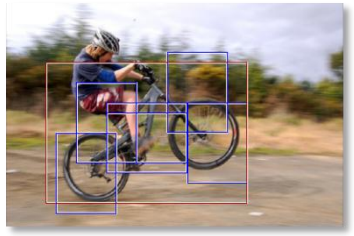
We generally have 13 classes.



1. Low-level vision (2-3 classes)
   - image processing, edge detection, feature detection, cameras, image formation



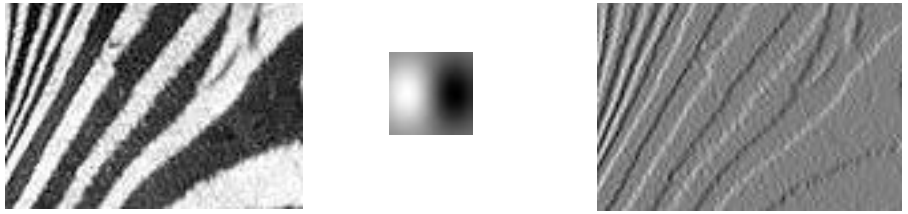
2. Geometry & appearance (3-4 classes)
   - projective geometry, stereo, structure from motion, optimization, lighting & materials



3. Recognition & generative models (6-8 classes)
   - object classification, deep learning, diffusion models

1. Low-level vision

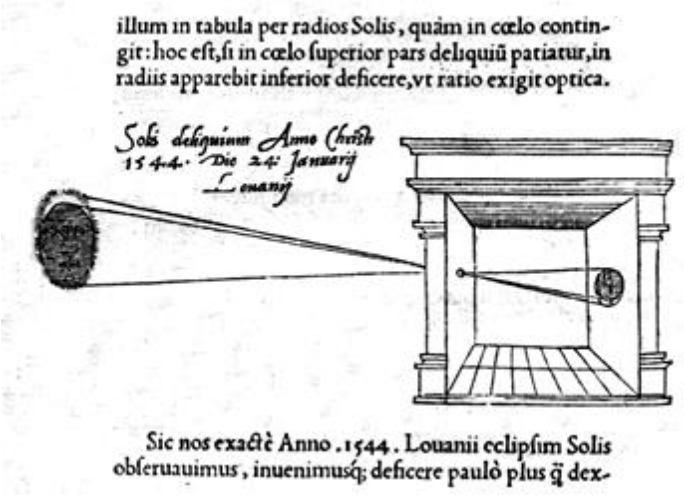
- Basic image processing and image formation



Filtering, edge detection



Feature extraction



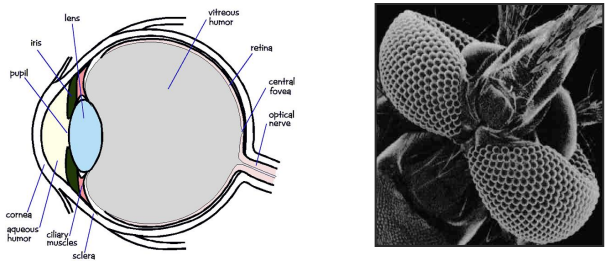
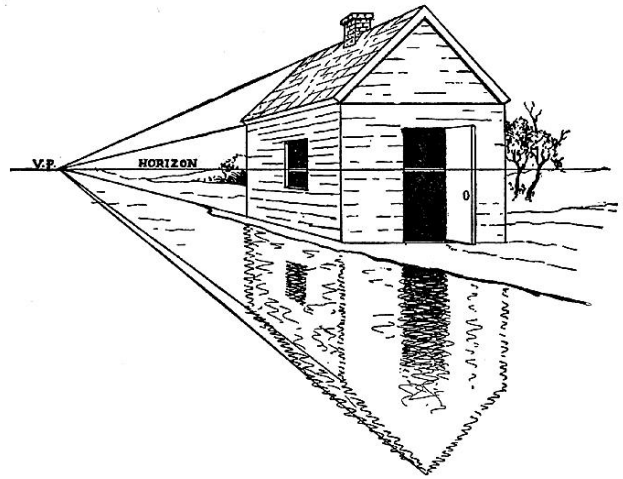


Image formation

2. Geometry & appearance



Projective geometry

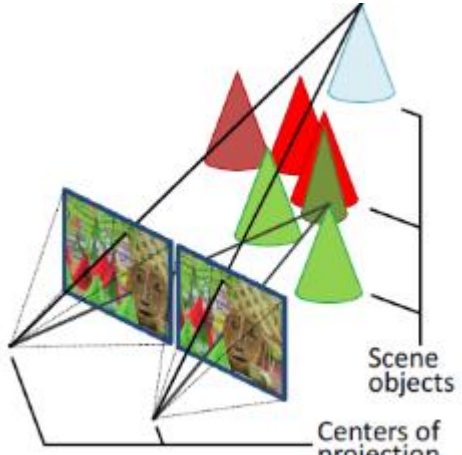
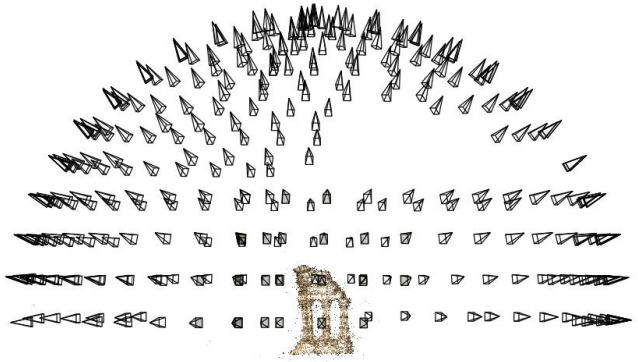


Image credit: IDS Imaging

Stereo vision



Multi-view stereo



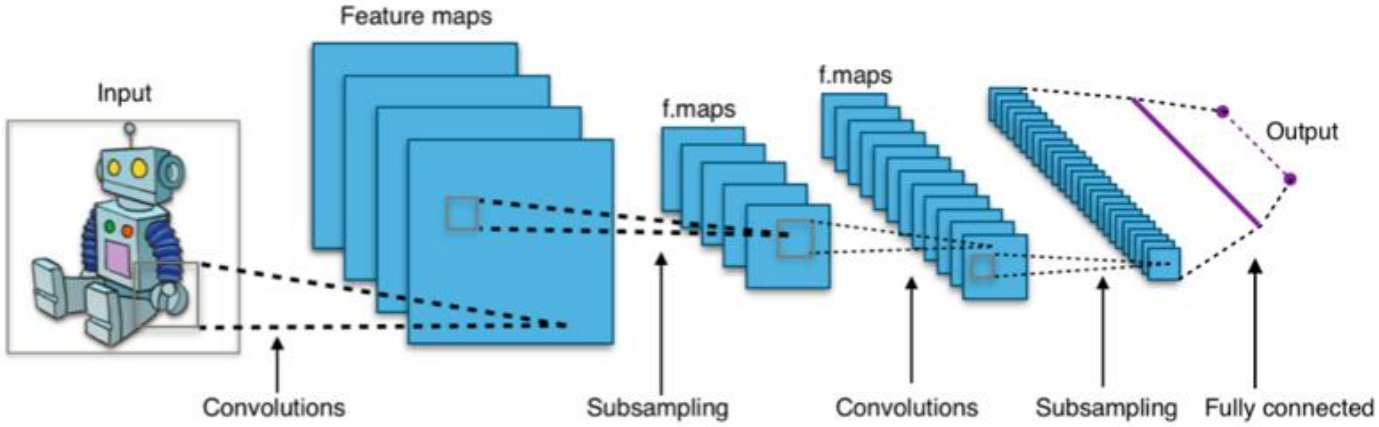
Structure from motion

3. Recognition & Generative Models



“dog”

Image classification



Convolutional Neural Networks



“a class watching a computer vision lecture”

Image generation

Lectures

- Lectures will be held in person in AC5-514
- If there is an instance where you cannot attend the class, please reach out to the instructor for approval
- No conversations during class, please!

Grading

- One midterm (between the 2nd and 3rd sections), and one final exam (in class)

- Grade breakdown (subject to minor tweaks):
  - Assignment : 40% = (Midterm: 30% + In-Class Quiz 10%)
  - Programming projects (2 projects): 10%
  - Final exam: 50%

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Academic Integrity

- Assignments will be done in pairs
- Please do not leave any code public on GitHub (or the like) at the end of the semester!
- If you use ChatGPT (or CoPilot, or similar) on coding assignments, you must disclose that with your submission
  - **BUT**: We advise you to do all coding yourself.

Questions?

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